

(02) RING PENDULUM**Aim**

Evaluation of methods for measuring the time period of a pendulum and determination of the acceleration due to gravity using hoops of different diameters.

Materials and Apparatus

Hoops of various diameters, stopwatch, stands, two reference pin, piece of thread, vernier caliper, clamp, and knife edge

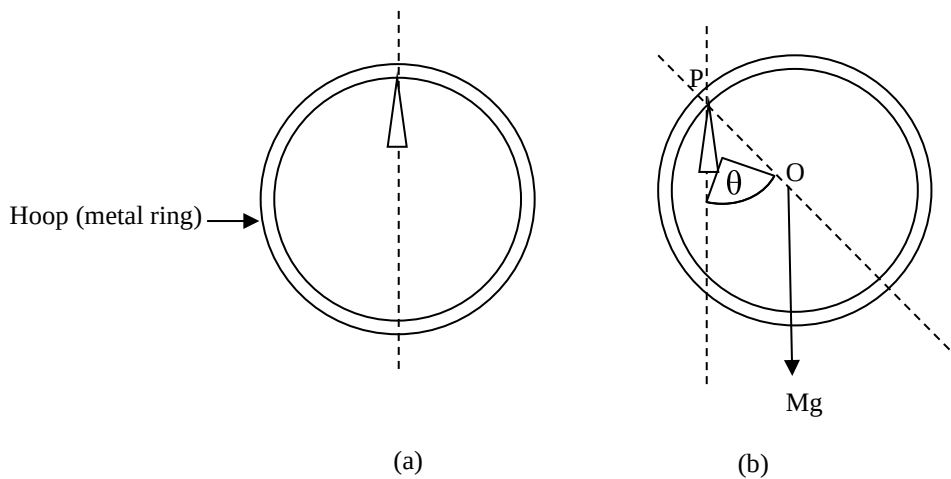
Theory**Oscillating Hoop as a Compound pendulum**

Fig. 1: Schematic diagram of the (a) hoop balanced on a knife edge (b) oscillating hoop with respect to P

Moment of inertia of a uniform hoop with respect to its center of mass is given by,

$$I = MR^2$$

where,

M : Mass of the hoop

R : Radius of the hoop

When the hoop is oscillating from a knife edge, the period of oscillation is given by,

$$Mg \sin(\theta) \times R = -I\ddot{\theta} \quad (1)$$

From parallel axis theorem, moment of inertia of the hoop with respect to the axis passing through point P,

$$I = I_c + MR^2$$

where,

I_c : Moment of inertia from the center of mass

$$I = 2MR^2 \quad (2)$$

From equations (1) and (2),

$$\ddot{\theta} = -\frac{g}{2R} \theta$$

Then the period of oscillation is given by,

$$T = 2\pi \sqrt{\frac{D}{g}} \quad (3)$$

where,

$$D = 2R$$

From equation (3),

$$T^2 = \frac{4\pi^2}{g} D$$

$\uparrow$ $\uparrow$
 y x
 $= m$

g can be obtained from the gradient of the graph of T^2 vs D

Procedure

Part I

- Arrange the apparatus as shown in Figure 1 (**Select one ring**).
- Measure the time taken for 1 oscillation using the stopwatch and take 10 such readings.
- Measure the time taken for 10 oscillations using the stopwatch and take 10 such readings.
- Calculate the average value, standard deviation (error associated with each measurement) for each set of readings, and standard error (error associated with the average) for the time taken for one oscillation.
- By comparing the fractional errors, identify the best approach to determine the period.

Note:

- The least count of the electronic stopwatch is usually 0.01 s.
- In general, humans are able to measure time with a precision of only about 0.1 seconds (when starting and stopping a stopwatch)

Therefore, a human can't measure the time duration in this experiment with the stopwatch with the accuracy of 0.01 s.

Part II

a) Give a small oscillation to the hoop and measure the time duration for a suitable number of oscillations.

Q: What is the reason behind giving small oscillations to the hoop?

Q: How do you decide the number of oscillations to minimize the error appropriately?

Q: In order to decide the number of oscillations, which hoop you should select? (smallest or largest).

b) Repeat the above procedure for other hoops.

c) Measure the diameters of the hoops.

Q: Propose two methods to measure the diameters of the hoops.

d) Determine the acceleration due to gravity using a graphical method and estimate the error associated.